So, the age of the eldest son when the third child was born = (4 + 5) years = 9 years.

When the fourth child was born, the total age of all the family members = (15 × 6) years

= 90 years.

By the time, the fourth child was born, each of the five family members had grown by $\left(\frac{90-80}{5}\right) = 2$ years.

So, the age of the eldest son when the fourth child was born = (9 + 2) years = 11 years.

At present, the total age of all the 6 family members

= (16 × 6) years = 96 years.

By now, each one of the 6 members have grown by $\left(\frac{96-90}{6}\right) = 1$ year.

Hence, the present age of the eldest son

= (11 + 1) years = 12 years.

144. Let the fixed cost be ₹ x and the variable cost be ₹ y per boarder.

Then, $x + 25y = 700 \times 25 \Rightarrow x + 25y = 17500$ ...(*i*)

$x + 50y = 600 \times 50 \Rightarrow x + 50y = 30000$ ...(*ii*)

Subtracting (*i*) from (*ii*), we get: $25y = 12500$ or $y = 500$.

Putting $y = 500$ in (*i*), we get: $x = 5000$.

∴ Total expenses of 100 boarders = ₹ (5000 + 500 × 100)

= ₹ 55000.

Hence, average expense = ₹ $\left(\frac{55000}{100}\right)$ = ₹ 550.

145. Let the daily wage of a man be ₹ x. Then, daily wage of a woman = ₹ $(x - 5)$.

Now, $600x + 400(x - 5) = 25.50 \times (600 + 400)$

⇔ $1000x = 27500$

⇔ $x = 27.50$.

∴ Man's daily wages = ₹ 27.50; Woman's daily wages = $(x - 5)$ = ₹ 22.50.

146. Let the required mean score be x. Then,

$20 \times 80 + 25 \times 31 + 55 \times x = 52 \times 100$

⇔ $1600 + 775 + 55x = 5200$

⇔ $55x = 2825 \Leftrightarrow x = \frac{565}{11} \approx 51.4$.

147. Let the number of students in batches Y and Z be $6x$ and $7x$ respectively, and the number of students in batch X be y.

Then, $72y + 60 \times 6x = 69(6x + y)$

⇒ $72y + 360x = 414x + 69y$

⇒ $3y = 54x \Rightarrow y = 18x$.

∴ Required average = $\frac{72 \times 18x + 60 \times 6x + 50 \times 7x}{18x + 6x + 7x}$

$= \frac{1296 + 360 + 350}{31} = \frac{2006}{31} = 64.7$.

148. Let the total number of workers be x. Then,

$8000x = (12000 \times 7) + 6000(x - 7)$

⇔ $2000x = 42000$

⇔ $x = 21$.

149. Let the number of girls be x. Then, number of boys = $(600 - x)$.

Then, $\left(11\frac{3}{4} \times 600\right) = 11x + 12(600 - x)$

⇔ $x = 7200 - 7050$

⇔ $x = 150$.

150. Let the number of boys and girls in the class be $3x$ and x respectively. Let the average score of the girls be y.

Then, $3x(A + 1) + xy = (3x + x)$ A

⇒ $3(A + 1) + y = 4A$

⇒ $y = A - 3$.

151. Let the number of Mechanical Engineering graduates be M and the number of Electronics Engineering graduates be E.

Then, $2.45M + 3.56E = 3.12(M + E)$

⇒ $2.45M + 3.56E = 3.12M + 3.12E$

⇒ $0.44E = 0.67M$

⇒ $\frac{M}{E} = \frac{0.44}{0.67} = \frac{44}{67}$.

Since the ratio 44 : 67 is in its simplest form, so least number of Electronics graduates = 67.

152. Let the ratio be $k : 1$. Then,

$k \times 16.4 + 1 \times 15.4 = (k + 1) \times 15.8$

⇔ $(16.4 - 15.8)k = (15.8 - 15.4)$

⇔ $k = \frac{0.4}{0.6} = \frac{2}{3}$.

∴ Required ratio = $\frac{2}{3} : 1 = 2 : 3$.

153. Let the number of students with work experience be x and those without work experience be y.

Then, $18000x + 12000y = 16000(x + y)$

⇒ $2000x = 4000y$

⇒ $\frac{x}{y} = \frac{2}{1}$.

∴ Percentage of students with work experience

$= \left(\frac{2}{3} \times 100\right)\% = 66.67\%$.

Percentage of students without work experience

= (100 − 66.67)% = 33.33%.

154. Let x ml of kerosene be there in 1 litre mixture.

Then, quantity of petrol in 1 litre mixture = $(1000 - x)$ ml.

∴ $\frac{40}{1000}(1000 - x) + \frac{18}{900}x = 38$

⇒ $\frac{x}{25} - \frac{x}{50} = 2 \Rightarrow \frac{x}{50} = 2$

⇒ $x = 100$.

So, 1 litre mixture has 900 ml petrol and 100 ml kerosene.

Cost of 1 litre petrol = ₹ 30.

Cost of 1 litre kerosene = ₹ $\left[\left(1 - \frac{2}{3}\right) \times 30\right]$ = ₹ 10.

Cost of 1 litre mixture = ₹ $\left(\frac{30}{1000} \times 900 + \frac{10}{1000} \times 100\right)$ = ₹ 28.

∴ Additional amount earned by pump-owner

= ₹ (30 − 28) = ₹ 2.